MATH 243: Discussion Worksheet 10

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**Topics:** 15.6 Triple Integrals; 15.7 Triple Integrals in Cylindrical Coordinates; 15.8 Triple Integrals in Spherical Coordinates

Please put away all electronic devices, including cell phones and calculators.



1. Evaluate the triple integral $\iiint_E (xy + z^2)\, dV$ where $E = [0,2] \times [0,1] \times [0,3]$.
2. Evaluate $\iiint_E \frac{1}{x^3}\, dV$ where $E = \{(x,y,z) | 0 \le y \le 1, 0 \le z \le y^2, 1 \le x \le z+1\}$.
3. Evaluate $\iiint_E x\, dV$ where $E$ is the solid bounded by the paraboloid $x = 4y^2 + 4z^2$ and $x = 4$.
4. Find the volume of the solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$.
5. Express the triple integral $\iiint_E f(x,y,z)\, dV$ as an iterated integral in six different ways, where $E$ is the solid bounded by $y = 4 - x^2 - 4z^2$ and $y = 0$.
6. What surface does the equation $z^2 + r^2 = 9$ describe in cylindrical coordinates?
7. Evaluate $\iiint_E z\, dV$ where $E$ is enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
8. Find the volume of the solid that lies between the paraboloid $z = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 2$.
9. Find the volume of the solid in the first octant that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, and the planes $z = 0$ and $z = 4$.
10. Evaluate the iterated integral
$$\int_{-3}^{3} \int_0^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} \sqrt{x^2+y^2}\, dz\, dy\, dx$$
by changing to cylindrical coordinates.
11. Evaluate the triple integral $\iiint_E (x^2 + y^2)\, dV$ where $E$ is the solid region between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.
12. Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the $xy$-plane, and below the cone $z = \sqrt{x^2 + y^2}$.
13. Find the volume of the part of the ball $\rho \le a$ that lies between the cones $\phi = \frac{\pi}{6}$ and $\phi = \frac{\pi}{3}$.
14. Evaluate the following integrals by switching to spherical coordinates.

    (a) $\displaystyle\int_{-a}^{a} \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} \int_{-\sqrt{a^2-x^2-y^2}}^{\sqrt{a^2-x^2-y^2}} (x^2 z + y^2 z + z^3)\, dz\, dx\, dy.$

(b) $\displaystyle\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}}(x^2+y^2+z^2)^{3/2}\,dz\,dy\,dx.$

15. A solid lies inside the sphere $x^2+y^2+z^2=4z$ and outside the cone $z=\sqrt{x^2+y^2}$. Write a description of the solid using inequalities involving spherical coordinates.

## SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

**Cylindrical Coordinates.** In the cylindrical coordinate system, a point $P$ in space is represented by an ordered triple $(r,\theta,z)$, where $(r,\theta)$ gives the polar coordinates of the projection of $P$ onto the $xy$-plane, and $z$ is the same $z$-coordinate as in rectangular coordinates:

$$x=r\cos\theta,\qquad y=r\sin\theta,\qquad z=z.$$

If a solid $E$ can be described in cylindrical coordinates by

$$E=\{(r,\theta,z)\mid \alpha\le\theta\le\beta,\ h_1(\theta)\le r\le h_2(\theta),\ u_1(r,\theta)\le z\le u_2(r,\theta)\},$$

then

$$\iiint_E f(x,y,z)\,dV=\int_\alpha^\beta\int_{h_1(\theta)}^{h_2(\theta)}\int_{u_1(r,\theta)}^{u_2(r,\theta)} f(r\cos\theta,r\sin\theta,z)\,r\,dz\,dr\,d\theta.$$

Cylindrical coordinates can also be defined using polar coordinates in the $yz$-plane or the $xz$-plane instead of the $xy$-plane, and in each case the Jacobian factor is still $r$.

**Spherical Coordinates.** In the spherical coordinate system, a point $P$ in space is represented by an ordered triple $(\rho,\theta,\phi)$, where $\rho$ is the distance from the origin to $P$, $\theta$ is the same angle as in cylindrical coordinates, and $\phi$ is the angle between the positive $z$-axis and the line segment from the origin to $P$:

$$x=\rho\sin\phi\cos\theta,\qquad y=\rho\sin\phi\sin\theta,\qquad z=\rho\cos\phi.$$

If a solid $E$ can be described in spherical coordinates by

$$E=\{(\rho,\theta,\phi)\mid \alpha\le\theta\le\beta,\ h_1(\theta)\le\phi\le h_2(\theta),\ u_1(\theta,\phi)\le\rho\le u_2(\theta,\phi)\},$$

then

$$\iiint_E f(x,y,z)\,dV=\int_\alpha^\beta\int_{h_1(\theta)}^{h_2(\theta)}\int_{u_1(\theta,\phi)}^{u_2(\theta,\phi)} f(\rho\sin\phi\cos\theta,\rho\sin\phi\sin\theta,\rho\cos\phi)\,\rho^2\sin\phi\,d\rho\,d\phi\,d\theta.$$

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**Suggested Textbook Problems**
Section 15.6: 1-47, 57, 58
Section 15.7: 1-32
Section 15.8: 1-32,37-40,43-46